Advanced Power System Analysis

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Table of contents

2 Introduction **5**
- 2.1 First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al. . . . 5
- 2.2 Paper's Objectives . . . 5

3 Background Information **6**
- 3.1 Three-phase Branch Power Matrix . . . 6
- 3.2 Derive the Magnitude of the Voltage . . . 7
  - 3.2.1 Solving for the left hand side of Equation 3.2 . . . 7
  - 3.2.2 Solving for the right hand side of Equation 3.2 . . . 7

4 Equations Derivation **9**
- 4.1 Deriving the Real and Reactive Power Equations . . . 10
  - 4.1.1 Decomposing the Losses term in equation Equation 4.1 . . . 10
- 4.2 Deriving the Voltage Equation: . . . 11
- 4.3 Three-phase Unbalanced System . . . 13
  - 4.3.1 Derive the current equations . . . 14

5 Motivation **16**
- 5.1 State of the Art . . . 16
- 5.2 Research Gap . . . 17
- 5.3 Research Goal . . . 17
- 5.4 Research Objectives . . . 17
- 5.5 Research Questions . . . 18
- 5.6 Scope and Assumptions . . . 18
- 5.7 Expected Contributions . . . 18
- 5.8 References . . . 19

6 Goal **21**
- 6.1 Setup . . . 21
  - 6.1.1 Initial Case . . . 21
  - 6.1.2 Sweep the voltage at Node 4 . . . 23
  - 6.1.3 Answering the first question . . . 24

7 Introduction **28**

References 29

1

2 Introduction

In this document, I review two papers. The first papers introduces the derivation is for three-phase, balanced systems. The second derviation is for three-phase unbalanced systems.

2.1 First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al.

2.2 Paper's Objectives

1- How can we change the network configuration to reduce resistive losses? 2- How can the load be distributed such that no overloads occur on the T-Lines?

3 Background Information

NOTE: This file contains supporting matrix algebra and the fuller multiphase branch-flow formulation of Gan and Low (Gan and Low 2014). The working equations are in `ch2_voltage_reactive_equations.qmd`.

3.1 Three-phase Branch Power Matrix

Recall that $\bar{\mathbf{V}}_i$ and $\bar{\mathbf{I}}_{ij}$ are 3×1 column vectors. We previously used the per-phase complex power vector:

$$\bar{\mathbf{s}}_{ij} = \begin{bmatrix} P_{ij}^a + jQ_{ij}^a \\ P_{ij}^b + jQ_{ij}^b \\ P_{ij}^c + jQ_{ij}^c \end{bmatrix}$$

Gan and Low also use the full branch power matrix:

$$\bar{\mathbf{S}}_{ij} = \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \tag{3.1}$$

Expanding the matrix multiplication:

$$\bar{\mathbf{S}}_{ij} = \begin{bmatrix} \bar{V}_i^a (\bar{I}_{ij}^a)^* & \bar{V}_i^a (\bar{I}_{ij}^b)^* & \bar{V}_i^a (\bar{I}_{ij}^c)^* \\ \bar{V}_i^b (\bar{I}_{ij}^a)^* & \bar{V}_i^b (\bar{I}_{ij}^b)^* & \bar{V}_i^b (\bar{I}_{ij}^c)^* \\ \bar{V}_i^c (\bar{I}_{ij}^a)^* & \bar{V}_i^c (\bar{I}_{ij}^b)^* & \bar{V}_i^c (\bar{I}_{ij}^c)^* \end{bmatrix}$$

Note that the **diagonal elements** are the conventional complex powers for each phase:

$$\text{diag}(\bar{\mathbf{S}}_{ij}) = \bar{\mathbf{s}}_{ij}$$

The **off-diagonal elements** are cross-phase voltage-current products. They do not represent extra downstream branches or separate real/reactive phase powers. The full 3×3 matrix is useful for Gan and Low's branch-flow and optimization formulations, but it is not required just to substitute phase power into KVL.

3.2 Derive the Magnitude of the Voltage

Following the same approach as in Equation 4.20, we can obtain squared voltage magnitudes using the three-phase KVL equation:

$$\bar{\mathbf{V}}_j = \bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij}$$

To retain all cross-phase voltage products, multiply the equation on the right by its **Hermitian transpose**:

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})^H \tag{3.2}$$

The Hermitian transpose, denoted by H, means **conjugate transpose**; it does **not** mean extracting the diagonal. A 3×1 voltage vector times its 1×3 conjugate transpose produces a 3×3 matrix.

3.2.1 Solving for the left hand side of Equation 3.2

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = \begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix}_{3\times1} \begin{bmatrix} (\bar{V}_j^a)^* & (\bar{V}_j^b)^* & (\bar{V}_j^c)^* \end{bmatrix}_{1\times3} = \begin{bmatrix} |\bar{V}_j^a|^2 & \bar{V}_j^a(\bar{V}_j^b)^* & \bar{V}_j^a(\bar{V}_j^c)^* \\ \bar{V}_j^b(\bar{V}_j^a)^* & |\bar{V}_j^b|^2 & \bar{V}_j^b(\bar{V}_j^c)^* \\ \bar{V}_j^c(\bar{V}_j^a)^* & \bar{V}_j^c(\bar{V}_j^b)^* & |\bar{V}_j^c|^2 \end{bmatrix}_{3\times3} \tag{3.3}$$

Note that the diagonal elements are **squared** voltage magnitudes. The off-diagonal elements remain complex cross-phase voltage products.

Gan and Low define the voltage matrix as:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H \tag{3.4}$$

Similarly:

$$\bar{\mathbf{W}}_i = \bar{\mathbf{V}}_i\bar{\mathbf{V}}_i^H \tag{3.5}$$

3.2.2 Solving for the right hand side of Equation 3.2

We can apply the same Hermitian transpose to the right-hand side:

$$\bar{\mathbf{W}}_j = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i^H - \bar{\mathbf{I}}_{ij}^H\bar{\mathbf{Z}}_{ij}^H)$$

Voltage Eq. Note

$$(\bar{\mathbf{Z}}\bar{\mathbf{I}})^H = \bar{\mathbf{I}}^H \bar{\mathbf{Z}}^H$$

The multiplication order reverses because $(AB)^H = B^H A^H$. This is a property of the conjugate transpose, not merely a rule for fixing dimensions.

Expanding the right-hand side:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_i \bar{\mathbf{V}}_i^H - \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{V}}_i^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H \tag{3.6}$$

We previously defined the voltage matrix in Equation 3.5 and the branch-power matrix in Equation 3.1. Accordingly, we can define the current outer-product matrix as:

$$\bar{\mathbf{L}}_{ij} = \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H$$

Therefore, Equation 3.6 can be written as:

$$\boxed{\bar{\mathbf{W}}_j = \bar{\mathbf{W}}_i - \bar{\mathbf{S}}_{ij} \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{S}}_{ij}^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H} \tag{3.7}$$

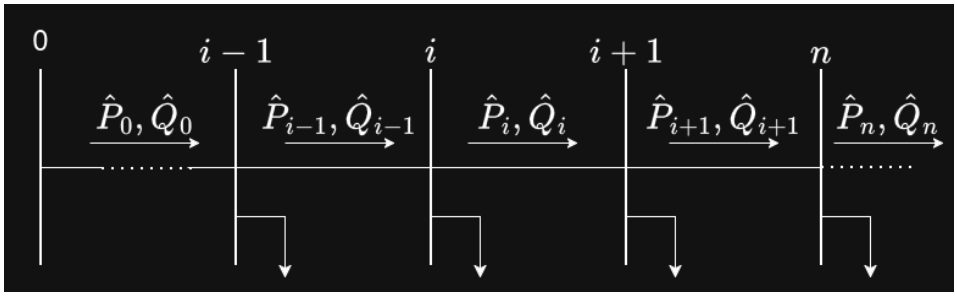
Note

Equation 3.7 is an exact squared-voltage **matrix** relationship derived from KVL. The term $\bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H$ is quadratic in line current; it is the outer product of the line voltage-drop vector with itself. Its diagonal is not the same as the real-power loss. For our immediate voltage–reactive-power work, the per-phase power equations and KVL are sufficient as starting equations, so we can keep this fuller derivation for background.

4 Equations Derivation

NOTE: Baran and Wu use the DistFlow equations for a balanced, single-phase equivalent radial distribution system. We use these equations as a starting point for our unbalanced three-phase work.

Consider the following single line diagram:



(a) ht

Figure 4.1: A single Line Diagram (SLD) for an n bus system. The loads are unknown for now. The complex power ($\bar{S}$) flowing between each bus is shown.

For each load, we consider $P_{L,n}$ and $Q_{L,n}$, indicating real and reactive power consumed at bus n. Positive load means consumption.

Each line has an impedance, $\bar{Z}_n = r_n + jx_n$. The sending-end complex power flow on each line is given by:

$$\bar{S_n} = P_n + jQ_n$$

For instance, for the line leaving bus $i+1$:

$$\bar{S}_{i+1} = P_{i+1} + jQ_{i+1}$$

4.1 Deriving the Real and Reactive Power Equations

I think it is easier to see this if we look at the power flow between two consecutive buses. Consider the power flow between buses n and $n+1$. We define P_n as the power leaving bus n and entering line $(n, n+1)$:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{P_{n,loss}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.1}$$

4.1.1 Decomposing the Losses term in equation Equation 4.1

$$P_{loss} = r|I|^2$$

At any given bus n:

$$P_{n,loss} = r_n|I_n|^2 \tag{4.2}$$

Recall:

$$\bar{S} = \bar{V}\bar{I}^* = P + jQ$$

Wherein the magnitude of the apparent power is given by:

$$|S| = |I|\ |V| = \sqrt{P^2 + Q^2}$$

Squaring both sides:

$$|S|^2 = |I|^2\ |V|^2 = P^2 + Q^2$$

Solving for the squared current magnitude:

$$|I|^2 = \frac{P^2 + Q^2}{|V|^2}$$

For the line leaving bus n, the squared current magnitude is:

$$|I_n|^2 = \frac{P_n^2 + Q_n^2}{|V_n|^2} \tag{4.3}$$

substituting the current equation Equation 4.3 into Equation 4.2 we get:

$$P_{n,loss} = r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2} \tag{4.4}$$

Substituting Equation 4.4 in the power balance between buses n and $n+1$ in Equation 4.1, we get:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.5}$$

> **Note**
>
> Equation 4.5 is the real power DistFlow Equation. This is the exact branch real-power balance for the single-phase equivalent line model, assuming the bus has one downstream line and no additional injection. For a branching feeder, we also account for every outgoing branch.

Similarly, the reactive power can be calculated in a similar manner:

$$\underbrace{Q_n}_{\text{Q leaving bus } n} = \underbrace{x_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Reactive power absorbed in the line}} + \underbrace{Q_{n+1}}_{\text{Q leaving bus } n+1} + \underbrace{Q_{L,n+1}}_{\text{Q consumed by bus } n+1 \text{ load}} \tag{4.6}$$

4.2 Deriving the Voltage Equation:

Applying KVL to buses, i and $i-1$ in Figure 4.1:

$$\bar{V}_{i-1} = \bar{Z}_{i-1}\ \bar{I}_{i-1} + \bar{V}_i \tag{4.7}$$

The current complex value can be solved for in a similar manner as in equation Equation 4.3:

$$\bar{V}_n\ \bar{I}_n^* = P_n + jQ_n$$

Taking the complex conjugate and solving for $\bar{I}_n$:

$$\bar{I}_n = \frac{P_n - jQ_n}{\bar{V}_n^*} \tag{4.8}$$

Substituting Equation 4.8 in Equation 4.7 & solving for $\bar{V}_i$:

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1} \tag{4.9}$$

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \frac{P_{i-1} - jQ_{i-1}}{\bar{V}^*_{i-1}}$$

We can then extend this to get only the magnitude of the voltage:

Recall $|a-b|^2 = (a-b)(a^*-b^*)$:

$$|\bar{V}_i|^2 = (\bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1})\ (\bar{V}^*_{i-1} - \bar{Z}^*_{i-1}\ \bar{I}^*_{i-1})$$

Multiplying both terms, we end up with:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2\ \Re\{\underbrace{\bar{V}^*_{i-1}\ \bar{I}_{i-1}}_{\substack{\text{Apparent}\\ \text{Power}\\ \text{Conjugate}}}\ \bar{Z}_{i-1}\} + |\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 \tag{4.10}$$

Recall $\bar{S}^*_{i-1} = \bar{V}^*_{i-1}\ \bar{I}_{i-1} = P_{i-1} - jQ_{i-1}$

So the middle term can be decomposed such that:

$$\bar{V}^*_{i-1}\ \bar{I}_{i-1}\ \bar{Z}_{i-1} = (P_{i-1} - jQ_{i-1})(r_{i-1} + jx_{i-1})$$

Multiplying and obtaining only the real part because we are trying to calculate the magnitude:

$$\Re\{\bar{Z}_{i-1}\ \bar{V}^*_{i-1}\ \bar{I}_{i-1}\} = r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1} \tag{4.11}$$

The last term in Equation 4.10 can be simplified further. Recall that $|\bar{Z}_{i-1}|^2 = r^2_{i-1} + x^2_{i-1}$ and $|\bar{I}_{i-1}|^2 = \frac{P^2_{i-1}+Q^2_{i-1}}{|\bar{V}_{i-1}|^2}$

$$|\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 = (r^2_{i-1} + x^2_{i-1})\ (\frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2}) \tag{4.12}$$

Substituting Equation 4.11 and Equation 4.12 in Equation 4.10:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}) + (r^2_{i-1} + x^2_{i-1})\ \frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2} \tag{4.13}$$

> **Note**
>
> Equation 4.13 is the Voltage Magnitude DistFlow Equation. This equation applies exactly to each branch under the paper's single-phase equivalent series-impedance model. It does not, by itself, describe mutual coupling in an unbalanced three-phase system.

4.3 Three-phase Unbalanced System

Using the same KVL we used before, Equation 4.9, we can set a system of equations for buses i and j such that i is upstream of j:

$$\begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} - \begin{bmatrix} \bar{Z}_{ij}^{aa} & \bar{Z}_{ij}^{ab} & \bar{Z}_{ij}^{ac} \\ \bar{Z}_{ij}^{ba} & \bar{Z}_{ij}^{bb} & \bar{Z}_{ij}^{bc} \\ \bar{Z}_{ij}^{ca} & \bar{Z}_{ij}^{cb} & \bar{Z}_{ij}^{cc} \end{bmatrix} \begin{bmatrix} \bar{I}_{ij}^a \\ \bar{I}_{ij}^b \\ \bar{I}_{ij}^c \end{bmatrix} \tag{4.14}$$

Note that the diagonal elements of $\bar{Z}_{ij}$ are self-impedances, and the off-diagonal elements are mutual impedances. This is a three-phase line model; a line with fewer phases uses the appropriate reduced impedance matrix. Neutral and transformer modeling must be handled consistently in the feeder model.

For any given phase, the voltage equation can be extracted from the above matrix form. For instance:

$$\bar{V}_j^a = \bar{V}_i^a - \bar{Z}_{ij}^{aa}\bar{I}_{ij}^a - \bar{Z}_{ij}^{ab}\bar{I}_{ij}^b - \bar{Z}_{ij}^{ac}\bar{I}_{ij}^c \tag{4.15}$$

We can rewrite the current terms in Equation 4.15 using complex power for each phase. At the sending end of line (i, j):

$$\begin{aligned} \bar{S}_{ij}^a &= \bar{V}_i^a(\bar{I}_{ij}^a)^* = P_{ij}^a + jQ_{ij}^a \\ \bar{S}_{ij}^b &= \bar{V}_i^b(\bar{I}_{ij}^b)^* = P_{ij}^b + jQ_{ij}^b \\ \bar{S}_{ij}^c &= \bar{V}_i^c(\bar{I}_{ij}^c)^* = P_{ij}^c + jQ_{ij}^c \end{aligned}$$

As a vector format, we can express the three complex power equations as:

$$\begin{bmatrix} \bar{S}_{ij}^a \\ \bar{S}_{ij}^b \\ \bar{S}_{ij}^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a(\bar{I}_{ij}^a)^* \\ \bar{V}_i^b(\bar{I}_{ij}^b)^* \\ \bar{V}_i^c(\bar{I}_{ij}^c)^* \end{bmatrix} = \begin{bmatrix} P_{ij}^a + jQ_{ij}^a \\ P_{ij}^b + jQ_{ij}^b \\ P_{ij}^c + jQ_{ij}^c \end{bmatrix} \tag{4.16}$$

> **Note**
>
> Equation 4.16 is a 3×1 vector of **per-phase complex powers**. It is not equal to a 3×3 matrix. The full branch-power matrix used by Gan and Low, including cross-phase products, is discussed in the background file. Mutual coupling is already present in Equation 4.14 through the off-diagonal impedances.

4.3.1 Derive the current equations

Consider the complex power for phase a between two buses:

$$\bar{S}_{ij}^{a} = \bar{V}_{i}^{a} (\bar{I}_{ij}^{a})^{*}$$

Take the complex conjugate of both sides:

$$\left(\bar{S}_{ij}^{a}\right)^{*} = (\bar{V}_{i}^{a})^{*} \bar{I}_{ij}^{a}$$

Solving for $\bar{I}_{ij}^{a}$:

$$\bar{I}_{ij}^{a} = \frac{\left(\bar{S}_{ij}^{a}\right)^{*}}{(\bar{V}_{i}^{a})^{*}}$$

Which can be rewritten as:

$$\bar{I}_{ij}^{a} = \frac{P_{ij}^{a} - jQ_{ij}^{a}}{(\bar{V}_{i}^{a})^{*}} \tag{4.17}$$

Solving for phases b and c using the same approach:

$$\bar{I}_{ij}^{b} = \frac{P_{ij}^{b} - jQ_{ij}^{b}}{(\bar{V}_{i}^{b})^{*}} \tag{4.18}$$

$$\bar{I}_{ij}^{c} = \frac{P_{ij}^{c} - jQ_{ij}^{c}}{(\bar{V}_{i}^{c})^{*}} \tag{4.19}$$

Equation 4.17, Equation 4.18, and Equation 4.19 can be used in our KVL equation Equation 4.14, such that:

$$\begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} - \bar{Z}_{ij} \begin{bmatrix} \dfrac{P_{ij}^a - jQ_{ij}^a}{(\bar{V}_i^a)^*} \\ \dfrac{P_{ij}^b - jQ_{ij}^b}{(\bar{V}_i^b)^*} \\ \dfrac{P_{ij}^c - jQ_{ij}^c}{(\bar{V}_i^c)^*} \end{bmatrix} \tag{4.20}$$

Equation 4.20 expresses the receiving-end voltage phasors as a function of the sending-end voltage phasors, branch impedance, and phase real and reactive power flows (Gan and Low 2014):

$$\boxed{\bar{\mathbf{V}}_j = \mathbf{f}(\bar{\mathbf{V}}_i, \bar{\mathbf{Z}}_{ij}, \mathbf{P}_{ij}, \mathbf{Q}_{ij})}$$

> **Note**
>
> This is an **exact branch relationship**, not yet an explicit voltage-magnitude sensitivity to inverter reactive power injection. To connect an inverter at bus k to voltages throughout the feeder, we must also use network current/power balance and the inverter injection convention. Gan and Low's squared-voltage matrix derivation is in the background file for reference.

5 Motivation

Distribution voltage is conventionally held near nominal along the primary feeder, typically within ±5%, so that every customer receives a service voltage inside ANSI C84.1 Range A (114–126 V). The limit, however, applies at the customer service point, not on the primary. If residential inverters can correct the voltage locally at each household, the primary feeder voltage no longer has to be chosen to protect the worst-placed customer. It becomes a decision variable: it can be lowered for conservation voltage reduction (CVR) or raised to reduce losses, while inverters hold each household within its service limit.

5.1 State of the Art

- **CVR with smart inverters.** Smart inverters have been coordinated with tap changers, regulators and capacitors to achieve CVR on three-phase unbalanced feeders (**jha2019bilevel?**; **zhang2019distributed?**), including integrated primary-secondary networks with voltage-dependent loads (**zhang2021primarysecondary?**). In these studies inverters *support* CVR, while the primary voltage stays within its conventional band.
- **Grid-edge voltage control.** Dedicated utility-owned devices on the secondary side of service transformers regulate customer voltage so that the primary can be lowered further. Field deployments report up to 5% CVR (**moghe2016gridedge?**), limiting-node voltage increases of 0.88–2.04% (**jha2023gridedge?**), and energy savings up to 4.7% when combined with smart inverters (**padullaparti2023cvr?**).
- **Volt-VAR curve design.** Per-inverter Volt-VAR rules have been optimized for single-phase feeders (**murzakhanov2024optimal?**), unbalanced multiphase feeders (**gupta2025regulating?**), and a single utility-scale battery (**ingebrigtsen2026optimizing?**). All aim to keep voltages near nominal rather than to compensate a deliberately shifted primary.
- **Cross-phase coupling.** On unbalanced four-wire networks, reactive power injected on one phase changes the voltages on the other phases (**wang2020interphase?**; **dalal2024crossphase?**).
- **Siting and sizing.** Methods exist to choose how many voltage-support units are needed and where, but for balanced networks (**giannitrapani2017optimal?**) or PV at medium voltage (**chen2022optimal?**).

5.2 Research Gap

Existing work either regulates customer voltage at the grid edge with dedicated utility devices, or uses residential inverters as auxiliary support while the primary voltage remains within its conventional band. No study has treated the primary feeder voltage as a free decision variable whose feasible range is set by the corrective capability of customer-owned residential inverters.

In particular, it is not known:

1. how far the primary voltage can be shifted, up or down, before participating inverters can no longer hold every household within its service-voltage limit;
2. how this range depends on the number, location and phase of participating inverters, and how it is reduced by cross-phase coupling, non-participating customers and battery operating state on unbalanced feeders;
3. whether the energy saved by operating the primary away from nominal outweighs the additional losses caused by the reactive power needed to compensate it; and
4. whether the required compensation can be delivered autonomously through location-dependent Volt-VAR settings.

We define the **decoupling range** as the set of primary-side voltage set-points for which the participating inverters can keep every household service voltage within its target limits.

5.3 Research Goal

To develop and evaluate a methodology for inverter-enabled grid-edge voltage regulation, in which residential inverters compensate their local service voltage so that the primary feeder voltage can be decoupled from customer voltage limits on unbalanced distribution networks.

5.4 Research Objectives

- **O1 — Quantify the decoupling range.** Develop a sensitivity-based model that determines the decoupling range as a function of inverter count, location, phase assignment and rating.
- **O2 — Identify the limiting factors.** Characterize how cross-phase coupling, non-participating customers and battery operating state (charging, discharging, idle) narrow or extend the decoupling range.
- **O3 — Evaluate the net benefit.** Determine the primary voltage set-point that maximizes net benefit across loading conditions, accounting for voltage-dependent load reduction, additional network losses, transformer loading and substation power factor.

- **O4 — Enable decentralized implementation.** Translate the required reactive power into location-dependent Volt-VAR settings (reference voltage, deadband, slope and reactive limits). Identify when autonomous operation is sufficient and when coordination is needed.

5.5 Research Questions

RQ1 — Decoupling feasibility: For a given set of residential inverters, what range of primary voltage set-points can be supported while keeping every household within its target service voltage, and how does this range scale with inverter count, location, phase assignment and rating?

RQ2 — Limiting factors: How do cross-phase coupling, non-participating customers and battery operating state constrain the decoupling range on unbalanced feeders?

RQ3 — Net benefit: Within the feasible range, which primary voltage set-point maximizes net energy benefit, after netting voltage-dependent load reduction against additional losses, and how does this optimum change with loading?

RQ4 — Decentralized implementation: Can the reactive power required at each location be encoded in Volt-VAR curves that hold household voltages at their targets without communication, and under what conditions does coordination become necessary?

5.6 Scope and Assumptions

- Unbalanced three-phase feeders with service transformers and single-phase residential connections. *To decide: test feeder (e.g., IEEE 8500-node or a PNNL taxonomy feeder) and whether secondaries are modelled as split-phase.*
- Customer-owned single-phase residential inverters (PV and battery) with reactive capability limited by IEEE 1547-2018.
- Voltage-dependent load models (ZIP or CVR factor), since CVR savings depend directly on them.
- Quasi-static time-series simulation over representative loading conditions.

5.7 Expected Contributions

1. The concept and quantification of a **decoupling range** for inverter-enabled grid-edge voltage regulation, and its dependence on inverter participation, location and phase.
2. An analysis of how **cross-phase coupling and battery operating state** limit grid-edge compensation on unbalanced feeders.

3. A **net-benefit assessment** identifying when shifting the primary voltage is worthwhile once added losses are included.
4. A method for deriving **location-dependent Volt-VAR settings** that implement the approach autonomously. Volt-VAR curves are the implementation of the methodology, not its primary objective.

5.8 References

- jha2019bilevel — Jha, Dubey et al., "Bi-Level Volt-VAR Optimization to Coordinate Smart Inverters With Voltage Control Devices," IEEE Trans. Power Syst., 2019. doi:10.1109/tpwrs.2018.2890613
- zhang2019distributed — Zhang, Dehghanpour and Wang, "Distributed CVR in Unbalanced Distribution Systems With PV Penetration," IEEE Trans. Smart Grid, 2019. doi:10.1109/tsg.2018.2880419
- zhang2021primarysecondary — Zhang, Guo et al., "Distributed Optimal Conservation Voltage Reduction in Integrated Primary-Secondary Distribution Systems," IEEE Trans. Smart Grid, 2021. doi:10.1109/tsg.2021.3088010
- moghe2016gridedge — Moghe, Tholomier et al., "Grid Edge Control: A new approach for volt-var optimization," IEEE PES T&D, 2016. doi:10.1109/tdc.2016.7519895
- jha2023gridedge — Jha, Fan et al., "Enhancing Conservation Voltage Reduction with Grid Edge Volt-VAr Control," IEEE Grid Edge, 2023. doi:10.1109/grid-edge54130.2023.10102726
- padullaparti2023cvr — Padullaparti, Baggu et al., "Conservation Voltage Reduction with Distributed Energy Resource Management System, Grid-Edge, and Legacy Devices," IEEE PESGM, 2023. doi:10.1109/pesgm52003.2023.10252669
- murzakhanov2024optimal — Murzakhanov, Gupta, Chatzivasileiadis and Kekatos, "Optimal Design of Volt/VAR Control Rules for Inverter-Interfaced Distributed Energy Resources," IEEE Trans. Smart Grid, 2024. doi:10.1109/TSG.2023.3279982
- gupta2025regulating — Gupta et al., "Regulating Voltage Magnitude and Phase Imbalance using DERs via Volt/VAR Control Rules," IEEE PowerTech, 2025. *(DOI to confirm)*
- ingebrigtsen2026optimizing — Ingebrigtsen, Bordin, Chiesa and Bakkejord, "Optimizing and implementing Volt-Var control in battery storage systems for voltage regulation in remote distribution networks," Discover Applied Sciences, 2026. doi:10.1007/s42452-025-08069-9
- wang2020interphase — Wang et al., "A Distributed Inter-Phase Coordination Algorithm for Voltage Control With Unbalanced PV Integration in LV Systems," IEEE Trans. Sustain. Energy, 2020. doi:10.1109/tste.2020.2970214
- dalal2024crossphase — Dalal et al., "Analyzing Cross-Phase Effects of Reactive Power Intervention on Distribution Voltage Control," IEEE PESGM, 2024. *(DOI to confirm)*
- giannitrapani2017optimal — Giannitrapani, Paoletti, Vicino and Zarrilli, "Optimal Allocation of Energy Storage Systems for Voltage Control in LV Distribution Networks,"

IEEE Trans. Smart Grid, 2017. doi:10.1109/TSG.2016.2602480
- chen2022optimal — Chen et al., "Optimal Placement of PV Smart Inverters With Volt-VAr Control," IEEE Systems Journal, 2022. *(DOI to confirm)*

6 Goal

We are trying to answer the following questions in this experiment: - Can an inverter at say, house 15, raise its voltage from (e.g., 0.92) up toward a target value (1.0 p.u.)? - If house 15 injects Q, how much do the voltages at the rest of the houses change? - How much reactive power is necessary to achieve a particular household voltage target? Is this amount within the inverter capability?

6.1 Setup

The GridLAB-D file is set to 15 houses, with a single substation transformer and distribution transformer. There are two nodes: upstream node and node 4. The upstream node is a swing bus node. Node 4 is the main node wherein all the loads connect to.

6.1.1 Initial Case

None of the houses have inverters in them. The voltagge behavior is depicted at each house, as well as the source node.

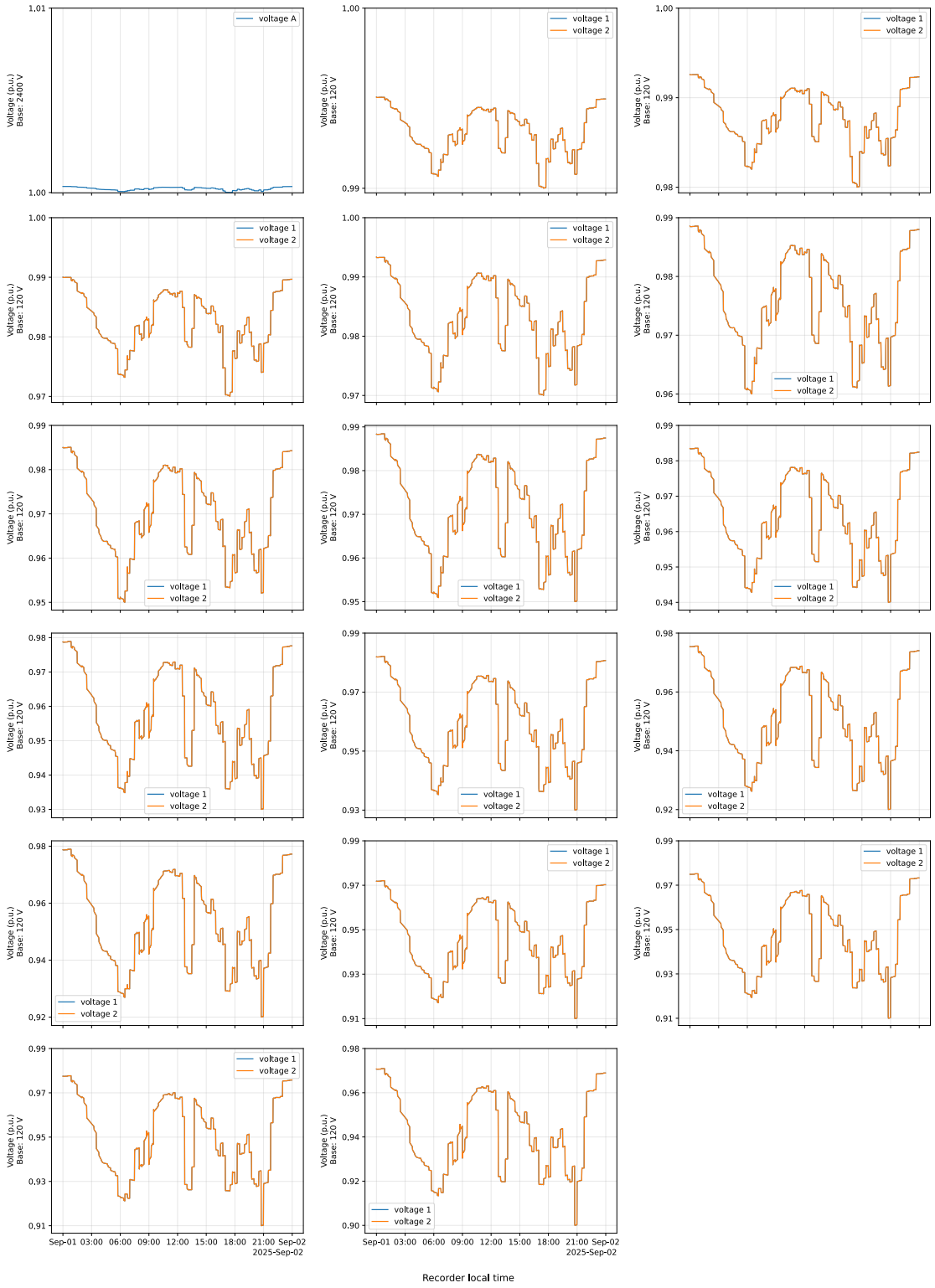


Figure 6.1: The voltage at each house, with a total of 15 houses. The first figure shows the voltage at Node 4, which is not a Swing node (obviously!)

Note that the voltage drops as the house is at a further distance from node 4.

6.1.2 Sweep the voltage at Node 4

When the voltage is swept at Node 4, the same voltage drop is depicted at different houses location.

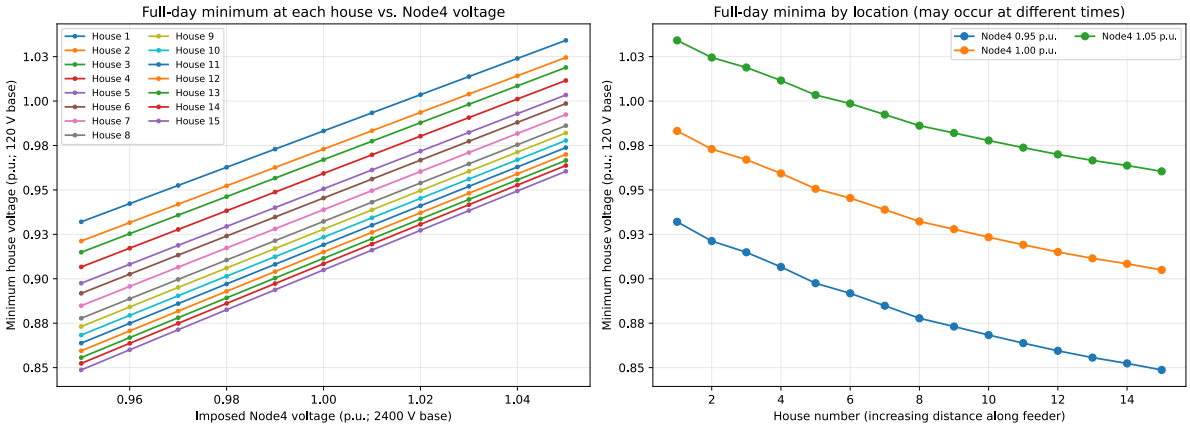


Figure 6.2: Voltage Sweeping Across Node4

Table 6.1: Simulation Results Data

node4_pu	house	min_pu	max_pu	min_timestamp	samples
0.95	10	0.868333	0.927508	2025-09-01 20:46:00 PDT	1439
0.95	11	0.863792	0.926525	2025-09-01 20:46:00 PDT	1439
0.95	12	0.859475	0.92555	2025-09-01 20:46:00 PDT	1439
0.95	13	0.85565	0.925008	2025-09-01 20:46:00 PDT	1439
0.95	14	0.852392	0.924492	2025-09-01 20:46:00 PDT	1439
0.95	15	0.8487	0.924225	2025-09-01 20:46:00 PDT	1439
0.95	1	0.932058	0.94515	2025-09-01 17:30:00 PDT	1439
0.95	2	0.921233	0.942225	2025-09-01 17:28:00 PDT	1439
0.95	3	0.914917	0.939467	2025-09-01 17:28:00 PDT	1439
0.95	4	0.906658	0.936867	2025-09-01 06:29:00 PDT	1439

Figure 6.2 demonstrates that the household voltage response depends on both the upstream voltage and the household's location along the feeder. Note that the distance of voltage lines in Figure 6.2 decreases consecutively. The current flow in the line supplying house 1 carries the combined current of many downstream loads. The line between house 1 and house 2 carries slightly less current, and so forth. From ohm's law:

$$\Delta\bar{\mathbf{V}} = \bar{\mathbf{Z}}\bar{\mathbf{I}}$$

If the line segments approximately equal impedances, the voltage drop across each segment generally decreases as we move downstream because the current decreases. Therefore:

$$|\Delta\bar{\mathbf{V}}_{1,2}| > |\Delta\bar{\mathbf{V}}_{13,14}| > |\Delta\bar{\mathbf{V}}_{14,15}|$$

To explain this further, let's look at all households at their lowest point, which is at 8:46 p.m.

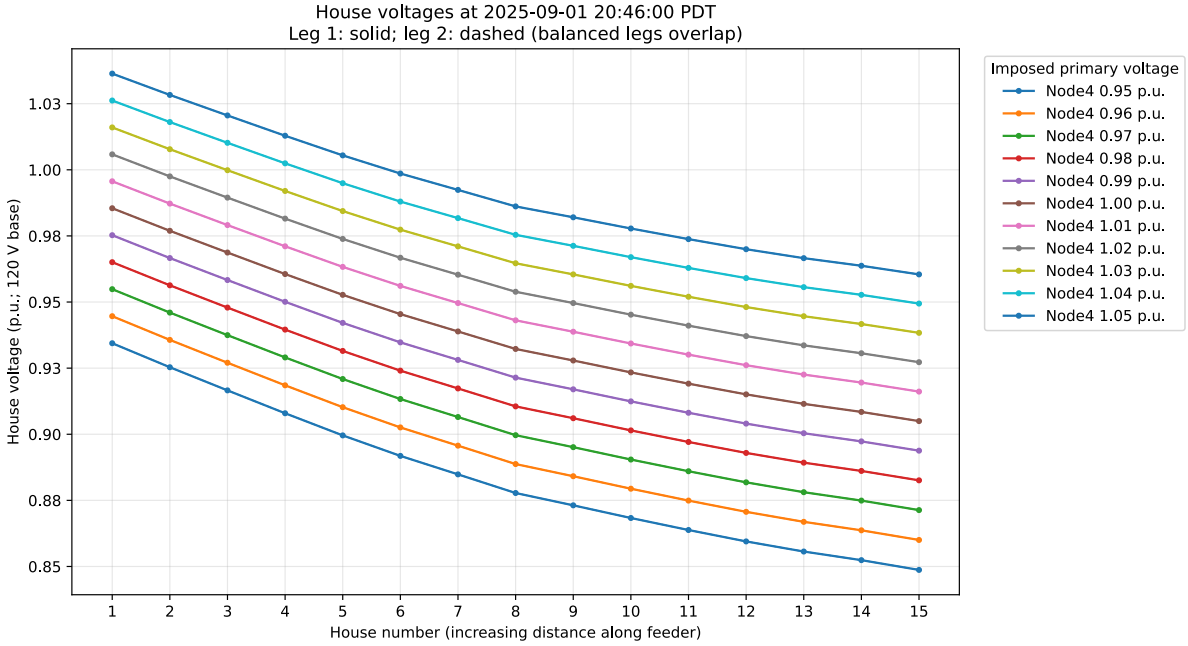


Figure 6.3: Voltage Sweep at 8:46 p.m. of all houses

The voltage behavior, as shown in Figure 6.3, is clearly nonlinear. And, the voltage drop distance between households decreases as we go down the node. This explains that the voltage trend is associated with the line impedance as well as the feeder loading, rather than a particular source voltage. This behavior is explain by:

$$\Delta\bar{\mathbf{V}} \approx \frac{RP + XQ}{\bar{\mathbf{V}}}$$

6.1.3 Answering the first question

The question that we try to answer in this section is:

- Can an inverter at say, house 15, raise its voltage from (e.g., 0.92) up toward a target value (1.0 p.u.)?

Not really. Figure 6.4 shows the behavior of a five kVA inverter installed at house 15. The inverter kept providing real power the whole day, resulting in $V_{max} = 0.98$. The constant injection of Q is due to the fact that the target voltage, $V_{target} = 1.00\ p.u.$ was never reached. Note that there is a slight change inthe head of the feeder (Node 4 at 2400 V). The change is almost zero. Also, the residential head (the secondary side of the transformer at 120 V) sees a low voltage change as well, at 0.006 V.

6.1.3.1 Determining the relationship between Q and the Voltage

We already know the relationship between the voltage and reactive power. From the Jacobian submatrix,

$$\bar{J}_4 = \frac{\partial \bar{Q}}{\partial |V|}$$

Sweeping the reactive power (Q) from $-5\ kVAr$ to $+5\ kVAr$ on a 5 kVA inverter at house 15 causes the voltage at that house to dropt first, because we are absorbing Q ($-Q$). Note that at the top plot of Figure 6.5 the house voltage goes from 0.88 p.u. to 0.92. However, doing the unreasonable thing and have that inverter inject reactive power (Q) all day long would improve the house15 voltage to 0.98 p.u., as shown in Figure 6.4.

The mid plot in Figure 6.5 shows the voltage at the distribution transformer in blue and the substation transformer in orange. The y-axis the voltage difference between the basecase and the Q sweep.

Note that the voltage deviation is worest at 20:46, so the Figure 6.5 is only running the simulation at that time.

The lower plot, showing the voltages at each house, as we sweep Q at the inverter at house 15. Note that the voltage difference between, for example, $+Q = 5\ kVAr$ and $Q = 0\ kVAr$ is larger at house 15 than at the first house. In fact, this difference decreases as we get closer to the head of the branch. Again, referring the Jacobian matrix, particularly, J_4, which describes the voltage/Q relationship:

$$\frac{\partial \bar{Q}_{15}}{\partial |V_i|} \quad \text{where, } i \ \in \{1, 2, 3, \cdots, 15\}$$

The U-shape curve of $V - Q$ curve can be obtained with more inverters installed and a well-optimized Q at each inverter (I think).

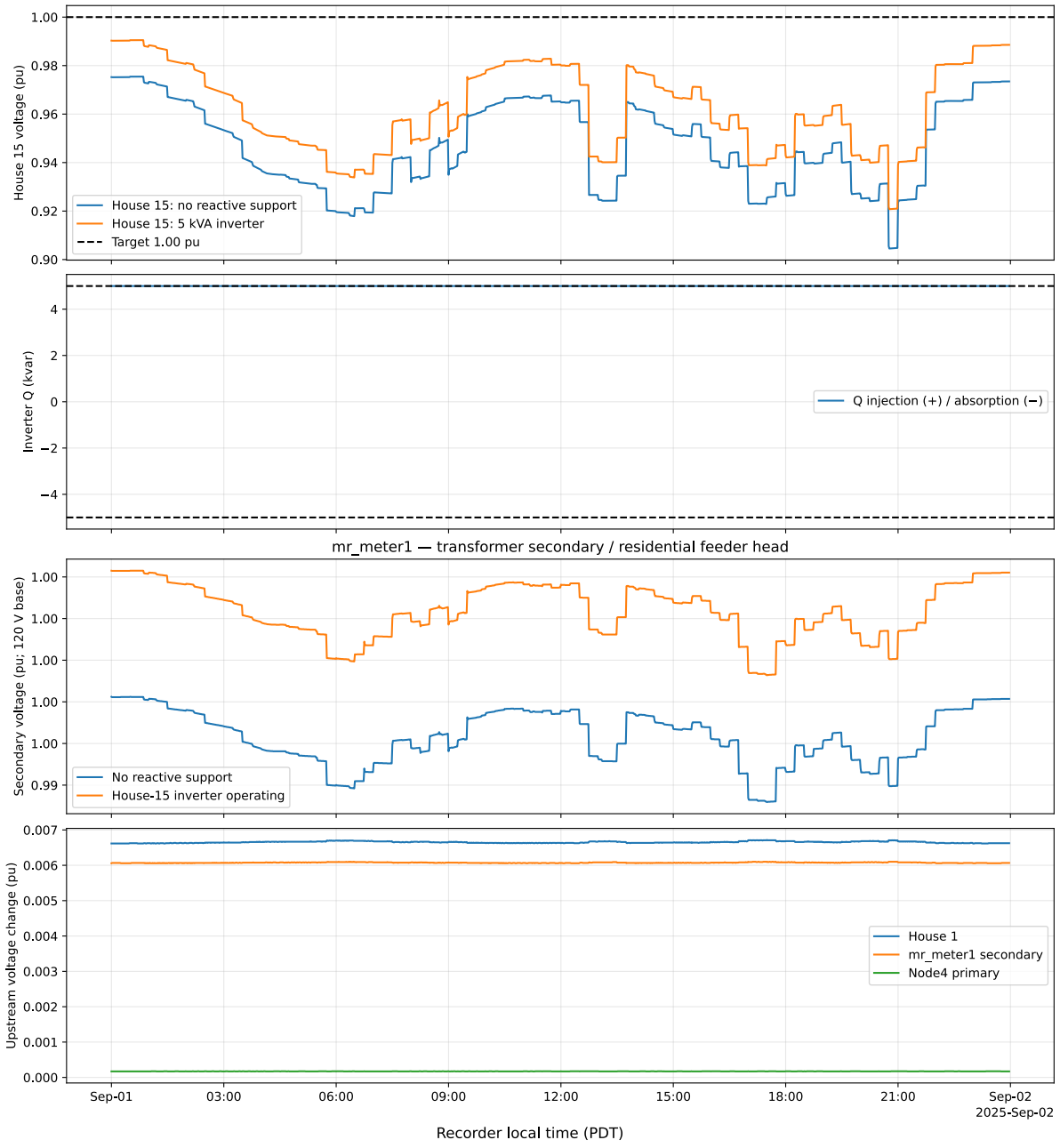


Figure 6.4: Inverter Operation at House 15

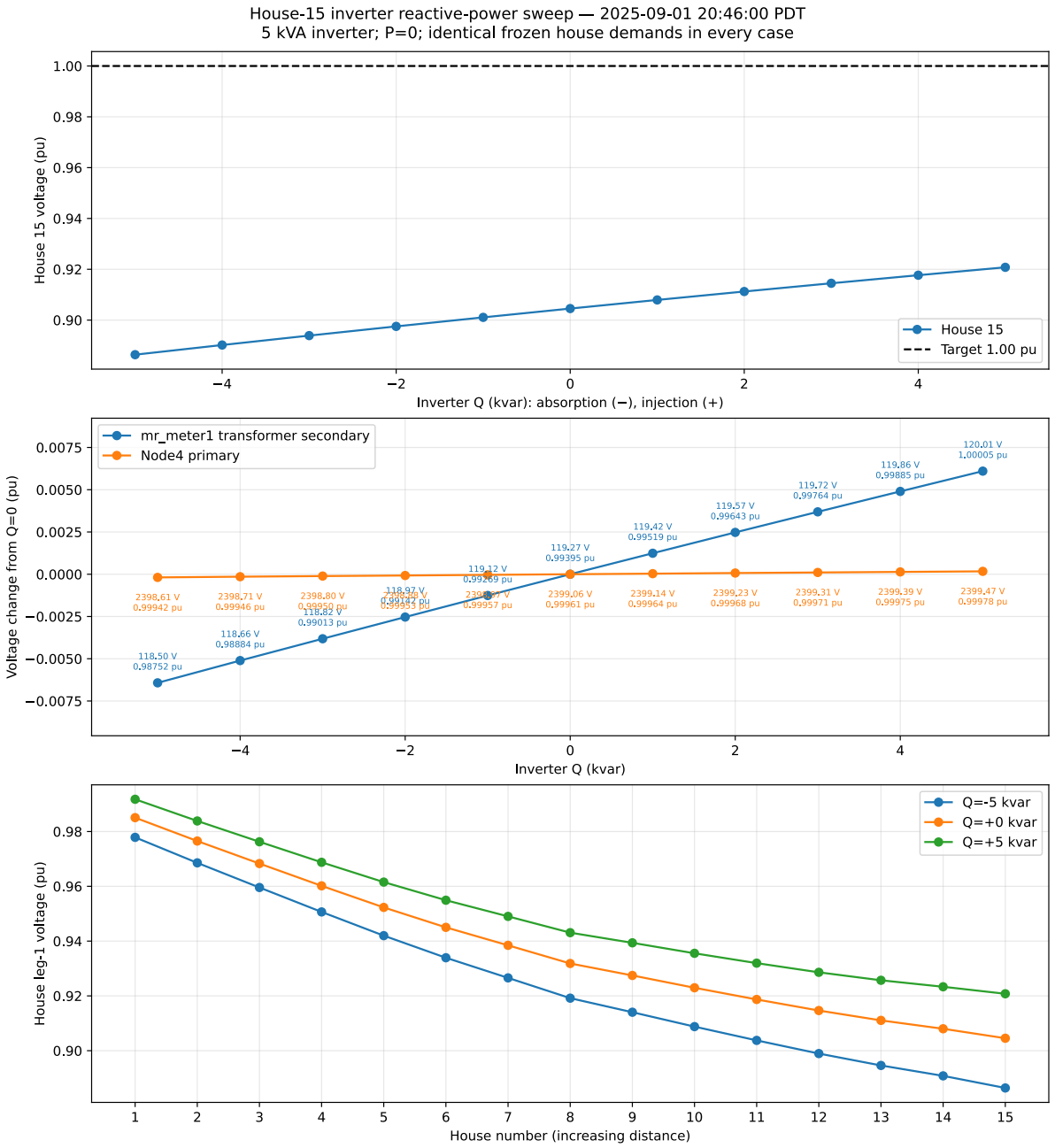


Figure 6.5: Inverter Operation at House 15

7 Introduction

References

Gan, Lingwen, and Steven H Low. 2014. "Convex Relaxations and Linear Approximation for Optimal Power Flow in Multiphase Radial Networks." *2014 Power Systems Computation Conference*, 1–9.